

The normal distribution appears to be a reasonable model of the behavior of certain random phenomena. It also is the limiting form of many other probability distributions, for example,

Theorem 1 A Poisson distributed random variable X with parameter λ ; then for fixed $a < b$

$$\Pr \left(a < \frac{X - \lambda}{\sqrt{\lambda}} < b \right) \rightarrow \Phi(b) - \Phi(a), \quad \text{as } \lambda \rightarrow \infty,$$

where $\Phi(\cdot)$ is the distribution function of standard normal.

Theorem 2 (De Moivre-Laplace limit theorem) A binomial distributed random variable X with parameters n and p ; then for fixed $a < b$

$$\Pr \left(a \leq \frac{X - np}{\sqrt{np(1-p)}} \leq b \right) \rightarrow \Phi(b) - \Phi(a), \quad \text{as } n \rightarrow \infty.$$

The usefulness of Theorems 1 and 2 rests in the approximations that they give. The normal distribution is also the limiting distribution in the famous *central-limit theorem*.

Theorem 3 (Central-limit theorem) For independent and identically distributed random variables $X_1, X_2, \dots, X_n$, from distribution $F_X(x)$ with mean μ_X and variance σ_X^2 , then for each z

the distribution function $F_{Z_n}(z)$ converges to $\Phi(z)$ as n approaches ∞ ,

where

$$Z_n = \frac{\bar{X}_n - E(\bar{X}_n)}{\sqrt{V(\bar{X}_n)}} = \frac{\bar{X}_n - \mu_X}{\sigma_X / \sqrt{n}}.$$

The key thing to note is that it does not make any difference what common distribution the $X_1, X_2, \dots, X_n, \dots$ have, as long as they have a mean and variance. A number of useful approximations can be garnered from the central-limit theorem, and they are listed as a corollary in the following.

Corollary 1 If $X_1, X_2, \dots, X_n$ are independent and identically distributed random variables with common mean μ_X and variance σ_X^2 , then

$$\begin{aligned} \Pr \left(a < \frac{\bar{X}_n - \mu_X}{\sigma_X / \sqrt{n}} < b \right) &\approx \Phi(b) - \Phi(a), \\ \Pr (c < \bar{X}_n < d) &\approx \Phi \left(\frac{d - \mu_X}{\sigma_X / \sqrt{n}} \right) - \Phi \left(\frac{c - \mu_X}{\sigma_X / \sqrt{n}} \right), \\ \Pr \left(r < \sum_{i=1}^n X_i < s \right) &\approx \Phi \left(\frac{s - n\mu_X}{\sqrt{n}\sigma_X} \right) - \Phi \left(\frac{r - n\mu_X}{\sqrt{n}\sigma_X} \right). \end{aligned}$$

These equations gives approximate values for the probabilities of certain events described in terms of averages or sums. The practical utility of the central-limit theorem is inherent in these approximations. The degree of approximation depends, of course, on the sample size n and on the particular distribution function of $X_i, F_X(x)$.

As for the effect of the particular distribution function $F_X(x)$ on the degree of approximation in the central-limit theorem, a function `simu.z()` written in R is designated to study this, along with sample size n , via simulation.

```

simu.z <- function(dstn="norm", n=31, mu, sigma, ..., R=1000, plot.it=TRUE) {
### Simulation study of z-statistic, namely, CLT
  dist <- sapply(c("d", "r"), function(k) get(paste(k, dstn, sep="")))
  if(missing(mu))
    mu <- integrate(function(x) x*dist$d(x, ...), -Inf, Inf)$value
  if(missing(sigma))
    sigma <- sqrt(integrate(function(x) (x - mu)^2*dist$d(x, ...),
      -Inf, Inf)$value)
  z <- sapply(1:R, function(dummy) {
    x <- dist$r(n, ...)
    (mean(x) - mu)/(sigma/sqrt(n))
  })
  digits <- round(log10(R))
  fournum <- fourNum(z)
  if(plot.it) {
    parm <- c(...)
    if(!is.null(parm)) {
      n.parm <- length(parm)
      parm <- paste(names(parm), parm, sep=" = ", collapse=", ")
    }
    else n.parm <- 0
    FourNum <- c(0, 1, 0, 0)
    layout(matrix(rep(1:4, rep(2,4)), 2, 4, byrow=TRUE)) # par(mfrow=c(2,2))
    hist(z, 20, ylim=c(0, dnorm(0)), freq=FALSE, col="gray", border="white",
      xlab=paste(R, "simulated z-values"),
      main="\nHistogram with Density Curve of N(0, 1)")
    curve(dnorm(x), col=2, add=TRUE)
    plot(1:10, type="n", ann=F, xaxt="n", yaxt="n", bty="n")
    mtext(expression("Statistic:"~z == frac(bar(x)[n] - mu,
      sigma/sqrt(n))), side=1, line=2)
    title(paste("\n\nDistribution parameter",
      ifelse(n.parm > 1, "s\n\n", "\n\n"),
      ifelse(is.null(parm), "(standard version)", parm),
      sep=""))
    position <- seq(7, by=-1.5, length=4)
    text(6.75, 8.5, "Sample statistics", cex=1.25, pos=2)
    text(rep(5,4), position,
      paste(c("mean", "variance", "skewness", "kurtosis"),
        "="), pos=2)
    text(rep(6.75,4), position, format(round(fournum, digits),
      digits=digits), pos=2)
    text(7, 8.5, "N(0, 1)", pos=4, cex=1.25)
    text(rep(7.5, 4), position, FourNum, pos=4)
    mtext(substitute("Sample from '~dn*'" Distribution of "~n == nn,
      list(dn=dstn, nn=n)),
      outer=TRUE, line=-1.5, cex=1.2)
    plot(sort(z), (1:R)/R, type="s", ylab="Probability",

```

```

      xlab="Simulated z-values", main="\nEmpirical CDF Plot")
curve(pnorm(x), col=2, add=TRUE)
abline(h=c(0,1), lty=3)
qqplot(qnorm(ppoints(R)), z,
      xlab=substitute("Quantiles of N(0, 1) Distribution", list(dd=df)),
      ylab="Simulated z-values",
      main="\nQ-Q Plot")
abline(0, 1, col=2)
layout(1) # par(mfrow=c(1,1))
}
return(round(fournum, digits))
}

```

where n (= 31, default) is sample size and R (= 1,000, default) is resampling size or resolution (the name we shall prefer).

Figure 1, generated with the code

```
> simu.z("pois", lambda = 2, mu = 2, sigma = sqrt(2))
```

shows the distribution of $(\bar{X}_n - \lambda)/\sqrt{\lambda/n}$ of simulated sample from $\text{Poisson}(\lambda = 2)$. The

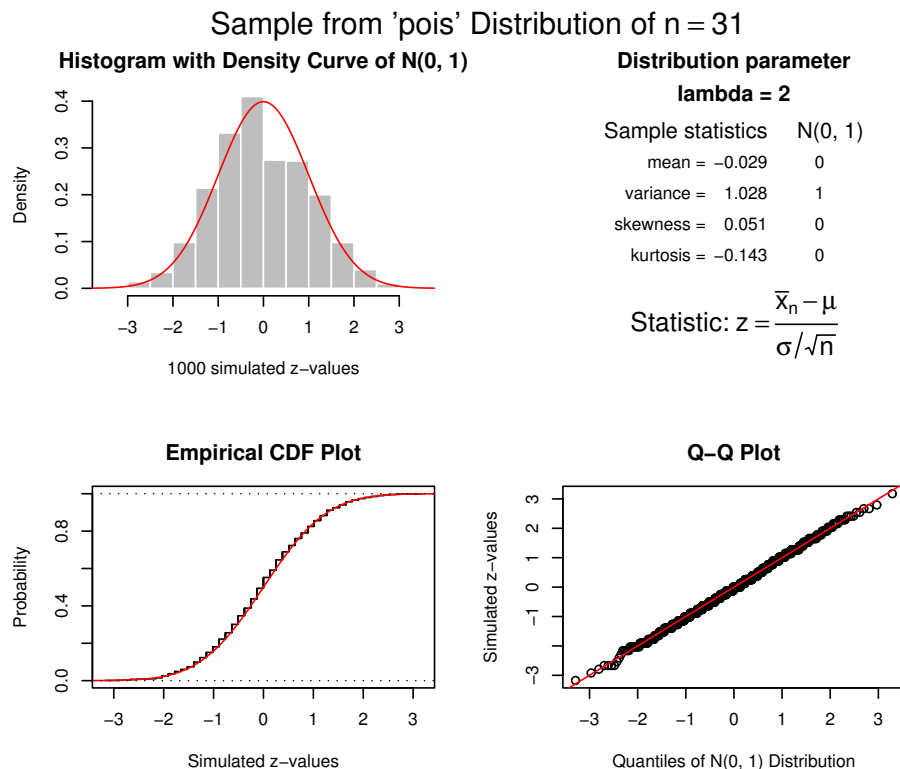


Figure 1: Simulated sample, of sample size 31, from $\text{Poisson}(\lambda = 2)$.

empirical (or sample) c.d.f. $F_n(x)$ of random sample $X_1, X_2, \dots, X_n$ is defined by

$$F_n(x) = \frac{\text{number of } X_i \text{ less than or equal to } x}{n} = \frac{1}{n} \sum_{i=1}^n I_{(-\infty, x]}(X_i).$$

noted that, in lower-left panel of Figure 1, we simply take those simulated z -values to make the empirical c.d.f. plot and hence the above n equals to the resolution $R (= 1,000)$. The Q-Q (quantile-quantile) plot of normal distribution is plotting

$$X_i \text{ versus } \Phi \left(\frac{\text{rank}(X_i) - 0.5}{n} \right),$$

for random sample $X_1, X_2, \dots, X_n$, and again, $n = R (= 1,000)$ in the resulting plot in lower-right panel of Figure 1. Notice, with the resolution 1,000, the closeness of the distribution of $(\bar{X}_n - \lambda)/\sqrt{\lambda/n}$, of sample size 31, from Poisson($\lambda = 2$) to that of standard normal.

Figure 2, generated with the code

```
> simu.z("binom", size = 10, prob = 0.1, mu = 1, sigma = sqrt(0.9))
```

shows the distribution of $(\bar{X}_n - mp)/\sqrt{mp(1-p)/n}$, of simulated sample, of sample size $n = 31$, from Binomial($m = 10, p = 0.1$). Noted that in R , the corresponding arguments to m and p are size and prob. With sample size 31, the closeness of the distribution of

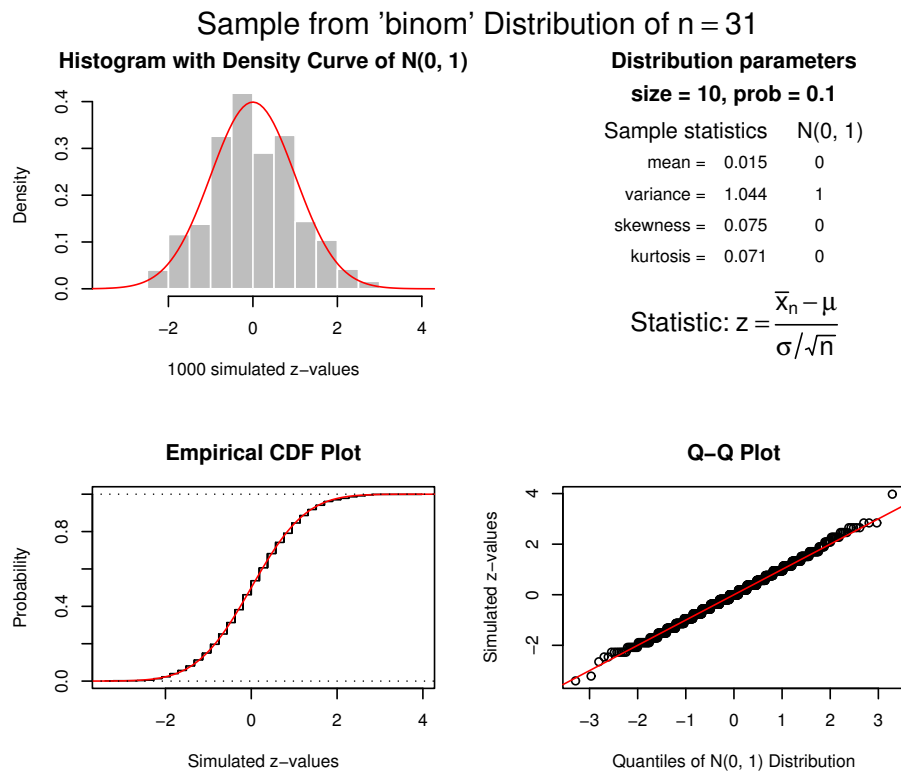


Figure 2: Simulated sample, of sample size 31, from Binomial($m = 10, p = 0.1$).

$(\bar{X}_n - mp)/\sqrt{mp(1-p)/n}$ for Binomial($m = 10, p = 0.1$) to that of standard normal, with resolution 1,000, is obvious.

Due to the difficulty in programming, in the use of `simu.z()`, the mean μ_X (`mu`) and the standard deviation σ_X (`sigma`) must be specified when the designed distribution is discrete. If left out without specified, μ_X and σ_X will be calculated, using numeric integration, for continuous distributions, however we might encounter

```
Error in integrate(function(x) (x - mu)^2 * dist$d(x, ...), -Inf, Inf) :
  non-finite function value
```

for example, `simu.z("logis")` for $\text{Logistic}(0, 1)$ distribution, and then we have to specify μ and σ manually. Noted that the mean and variance of $\text{Logistic}(\text{location}, \text{scale})$ are location and $(\pi \times \text{scale})^2/3$, respectively.

Figure 3, generated with the code

```
> simu.z("exp")
```

shows the distribution of $(\bar{X}_n - \theta)/(\theta/\sqrt{n})$ of simulated sample, with sample size $n = 31$, from $\text{Exponential}(\theta = 1)$, θ is the mean and the setting of exponential distribution in *R* is in terms of $\text{rate} = 1/\theta$ and its default is 1. Notice that, particularly both skewness

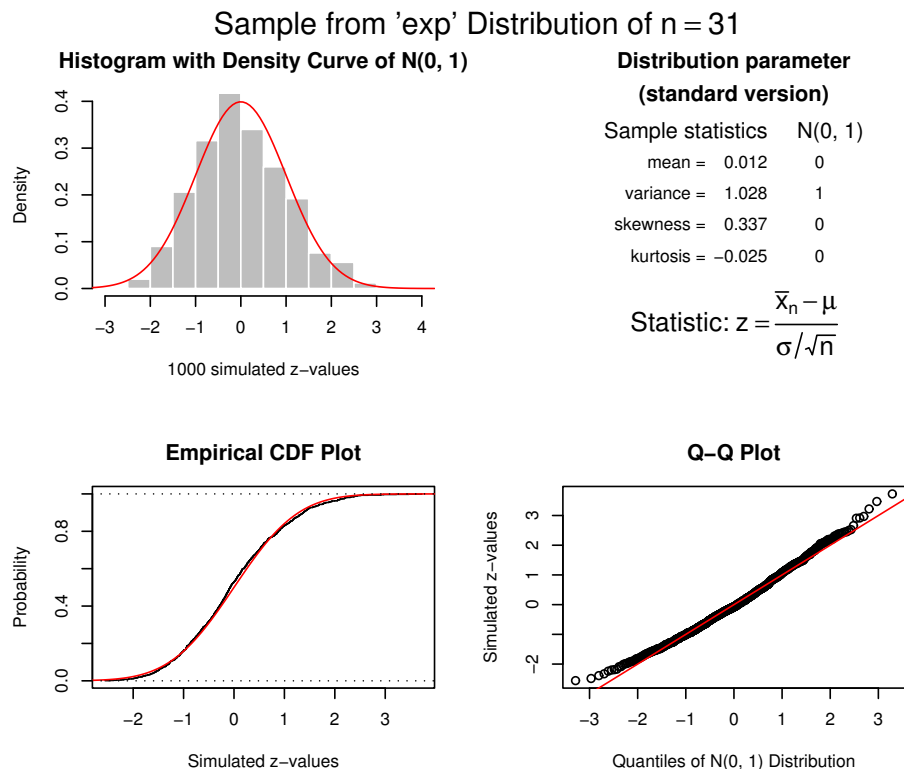


Figure 3: Simulated sample, of sample size 31, from $\text{Exponential}(\theta = 1)$.

and kurtosis not close to 0, the not quite close of the distribution of $(\bar{X}_n - \theta)/(\theta/\sqrt{n})$, for $\text{Exponential}(\theta = 1)$ of sample size 31 with resolution 1,000, to that of standard normal. To study the effect of sample size when the distribution is $\text{Exponential}(\theta = 1)$, we increased sample size and obtain the following:

```
> n <- c(5, 10, 30, 50, 100) * 10 + 1
> z.exp <- sapply(n, function(n) simu.z("exp", n=n, plot.it=FALSE))
> colnames(z.exp) <- paste("n", n, sep=" ")
> z.exp
```

	n=51	n=101	n=301	n=501	n=1001
mean	-0.002	0.006	0.004	0.027	0.002
variance	1.120	1.005	0.960	0.917	1.053
skewness	0.385	0.213	0.093	0.047	-0.020
kurtosis	0.619	0.077	-0.015	0.105	-0.062

Repeat this simulation several times and examine the resulting values, we shall have a clear picture about the central-limit theorem on the distribution $F_{Z_n}(z)$ when the underlying distribution is Exponential. (Question: Does this matter with different values of θ ?)

Besides those distributions mentioned in the above, we can use `simu.z()` to study central-limit theorem for other distributions, for example,

- `simu.z("unif")` for Uniform(0, 1) distribution,
- `simu.z("t", df = 5)` for t_5 -distribution,
- `simu.z("chisq", df = 10)` for χ^2_{10} -distribution,
- `simu.z("gamma", shape = 10, rate = 1)` for Gamma(shape = 10, rate = 1) distribution, or
- any user-written functions having density function `d_####()` and random value generating function `r_####()`.

With some modifications of `simu.z()`, similar functions for simulation study can be written for t -, χ^2 -, and F -statistics.

Noted that there are two functions called by `simu.z()`:

```
fourNum <- function (x) {
  mu <- mean(x)
  sigma <- sd(x)
  z <- scale(x, mu, sigma)
  c(mean = mu, variance = sigma^2, skewness = mean(z^3),
    kurtosis = mean(z^4) - 3)
}
```

to calculate four summary statistics and

```
ecdf <- function (x) {
  n <- length(x)
  function(t) {
    sapply(t, function(k) sum(x <= k)/n)
  }
}
```

to calculate empirical c.d.f. $F_n(x)$.

